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Professor Holden-Gouveia
Linux Administration
Due Date: 2/15/26
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SedLab

This document for the third lab named **SedLab,** contains screenshots of different sed commands; the screenshots are clearly labeled with each question it correlates to. Each screenshot shows the **command,** and the **results of the command** as what I believe to have been the best approach to each question. Each screenshot explains what the command does and why. I did my best to explain each question in the most detail as possible. I have cited all my sources used to finish each question, at the bottom of the command I wrote for each question.

Wrote these instructions one last time to better remember the steps that helped me for file sharing. **I will continue to use these steps in future labs that use file sharing.**

Question **1.**

```
  IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe5c:2b44

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

69 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update
New release '24.04.3 LTS' available.
Run 'do-release-upgrade' to upgrade to it.


Last login: Mon Feb  9 18:31:33 UTC 2026 on tty1
aavbox26@Ubuntu1:~$ sudo mkdir /mnt/sed
[sudo] password for aavbox26:
aavbox26@Ubuntu1:~$ cd sed
-bash: cd: sed: No such file or directory
aavbox26@Ubuntu1:~$ sudo mount -t vboxsf sed /mnt/sed
aavbox26@Ubuntu1:~$ ls/mnt/sed
-bash: ls/mnt/sed: No such file or directory
aavbox26@Ubuntu1:~$ ls /mnt/sed cat /mnt/sed/SedLab
ls: cannot access 'cat': No such file or directory
/mnt/sed/SedLab

/mnt/sed:
SedLab
aavbox26@Ubuntu1:~$ _
```

The very first thing to do is **"file sharing"** with the server. For this example I had a file named **SedLab,** and a shared folder named **"sed".** Make sure to **auto-mount** and **make machine permanent** options before continuing this lab assignment.

Important commands to use for file sharing in **Ubuntu.** The first command is **sudo mkdir /mnt/directory name.** In the example above it is written as **sudo mkdir /mnt/sed.** After writing this command the system will then ask for your **password** for the **user you created within the Ubuntu server** environment.

For this example, the username can be something like **aavbox26,** the password must contain **8 characters** with a mix of letters, numbers, and symbols but cannot be seen as **Ubuntu does not show anything for passwords.** The second command that should be written is **sudo mount -t vboxsf sed /mnt/sed.**

Example for commands:
https://help.ubuntu.com/community/VirtualBox/SharedFolders

The final command that should be written for file sharing should be **ls /mnt/directory name like myshare or sed,** then proceeding it in the same line is **cat /mnt/sed/SedLab or cat /mnt/directory name/folder name. Using these commands link the SedLab file to the Ubuntu server environment.**

Ls /mnt/myshare command shows the file in the **current directory,** as long as they are not hidden files. **Ls** stands for **list, mnt** stands for **mount, sed** is the **directory name** written in the example. The directory name can be whatever you choose and what is easy to remember for the current assignment or job task.

The (**cd) command** means **change directory, cd.. (cd, two dots)** means **going up one level from where the cursor currently is.** (**Cd..)** changes to **mnt then writing cd..** again takes you to the home**. Pwd** means **print working directory,** it tells you the path of the current directory you are in. After writing pwd and pressing enter, **it will show that you are located in the home, the /. The slash means root.**

Once you have the file in the directory, you should move it to your **linux server,** in this case **Ubuntu.** The command for this would be (**sudo mv** filename /home/username/directoryname).

Example: **sudo mv SedLab /home/aavbox26/sed.** To get into the directory you would write **(cd sed)** or **cd "directory name". Change directory, directory**

name. Sudo means superuser do, or substitute user do it gives regular users access to a command with administrative privileges of a user like root. Mv means move one file from one location to another. The destination for this command is **/home/aavbox26/sed. sed** is the directory name for the file named **SedLab.**

Source for command:

https://askubuntu.com/questions/465877/how-to-move-one-file-to-a-folder-using-terminal

```
aavbox26@Ubuntu1:/mnt/fileshare$ cp SedLab /home/
cp: cannot create regular file '/home/SedLab': Permission denied
aavbox26@Ubuntu1:/mnt/fileshare$ sudo cp SedLab /home/
aavbox26@Ubuntu1:/mnt/fileshare$ cd /home/
aavbox26@Ubuntu1:/home$ cat SedLab_
```

The first step to do this is to copy a file from a shared folder directory into the home directory. This is to prevent any errors that might occur when editing the file. The command we would write for this is, **cp SedLab /home/ . cp** means **copy**, then we have the filename followed by slash home directory slash.

```
Lizzie Bennett:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900

Nancy Drew:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900

Jo March:327-832-5728:3465 Mirlo Street, Peabody, MA 34756:10/2/65:35200

Victor Frankenstein:835-365-1284:454 Easy Street, Decatur, IL 75732:2/28/53:123500

Ephram Hardy:293-259-5395:235 CarltonLane, Joliet, IL 73858:8/12/20:56700

Meg Murry:834-938-8376:23445 Aster Ave., Allentown, NJ 83745:12/1/38:45000

Lucy Pevensie:385-573-8326:832 Ponce Drive, Gzary, IN 83756:12/1/46:268500

Hemione Granger:408-456-1234:4 Harvard Square, Boston, MA 02133:4/22/62:52600

Mary Poppins:846-836-2837:6937 Ware Road, Milton, PA 93756:9/21/46:43500

Sir Lancelot:837-835-8257:474 Camelot Boulevard, Bath, WY 28356:5/13/69:24500

Minerva McGonagall:408-233-8971:45 Rose Terrace, San Francisco, CA 92303:2/3/36:25000

Zippy Pinhead:834-823-8319:2356 Bizarro Ave., Farmount, IL 84357:1/1/67:89500

Dorothy Gale:923-835-8745:23 Wimp Lane, Kensington, DL 38758:8/31/69:126000

Popeye Sailor:156-454-3322:945 Bluto Street, NotHere, USA 29358:3/19/35:22350

Luna Lovegood:385-898-8357:38 Fife Way, Abilene, TX 39673:1/5/58:95600

Daenerys Targaryen:408-724-0140:1222 Oxbow Court, Sunnyvale, CA 94087:5/19/66:34200

Molly Weasley:387-827-1095:13 Uno Lane, Ashville, NC 23556:7/1/29:57000

Arya Stark:438-910-7449:8235 Maple Street, Wilmington, VM 29085:9/23/63:68900
aavbox26@Ubuntu1:/mnt/sed$ cat SedLab
```

The following command, **cat SedLab** was written to see the contents of the file named **SedLab** within the **Ubuntu directory** named **sed.** The **cat command** concatenates files. The command is used to view, read, and combine different file content.

```
aavbox26@Ubuntu1:/mnt/sed$ sed -n 's/Jo /Josephine /p' SedLab
Josephine March:327-832-5728:3465 Mirlo Street, Peabody, MA 34756:10/2/65:35200
aavbox26@Ubuntu1:/mnt/sed$ _
```

The following command shown in the screenshot above is....

sed -n 's/Jo /Josephine /p' SedLab. This command changes the only person's name **Jo to Josephine,** make sure to include the spaces in the command. For example, if you don't have the spaces it will change san jose into san josephine or any word containing **Jo. (p) stands for print,** it prints every instance of Jo or Josephine.

Sources :

https://www.grymoire.com/Unix/Sed.html#uh-1 **Substitute.**

https://www.grymoire.com/Unix/Sed.html#uh-15b **Pattern.**

Question **2.**

```
aavbox26@Ubuntu1:/home$ sudo sed -i -e '$d' SedLab
aavbox26@Ubuntu1:/home$ sudo sed -i -e '$d' SedLab
aavbox26@Ubuntu1:/home$ sudo sed -i -e '$d' SedLab
aavbox26@Ubuntu1:/home$ sudo sed -i -e '$d' SedLab
aavbox26@Ubuntu1:/home$ sudo sed -i -e '$d' SedLab
aavbox26@Ubuntu1:/home$
```

The following commands, delete the last **five lines** in the terminal, not in the actual file. The command was written in the **home directory** to avoid accidentally deleting more lines than needed or any other type of errors. The command written to delete five lines was sudo **sed -i -e '$d' SedLab. -i** edits a file directly and changes the original file, **-e** is used to tell sed to continue with the command, **$** the dollar sign is the end of the file, **d** deletes lines from a file, **SedLab** is the filename.

```
Holly Golightly:397-857-2735:74 Pine Street, Dearborn, MI 23874:3/28/45:245700

Ebenezer Scrooge:548-834-2348:583 Laurel Ave., Kingsville, TX 83745:10/1/35:58900

Hester Prynne:408-253-3122:123 Park St., San Jose, CA 04086:7/25/53:85100

Westley Pirate:284-758-2857:23 Edgecliff Place, Lincoln , NB 92743:7/25/53:85100

Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200

Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200

Lizzie Bennett:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900

Nancy Drew:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900

Jo March:327-832-5728:3465 Mirlo Street, Peabody, MA 34756:10/2/65:35200

Victor Frankenstein:835-365-1284:454 Easy Street, Decatur, IL 75732:2/28/53:123500

Ephram Hardy:293-259-5395:235 CarltonLane, Joliet, IL 73858:8/12/20:56700

Meg Murry:834-938-8376:23445 Aster Ave., Allentown, NJ 83745:12/1/38:45000

Lucy Pevensie:385-573-8326:832 Ponce Drive, Gzary, IN 83756:12/1/46:268500

Hemione Granger:408-456-1234:4 Harvard Square, Boston, MA 02133:4/22/62:52600

Mary Poppins:846-836-2837:6937 Ware Road, Milton, PA 93756:9/21/46:43500

Sir Lancelot:837-835-8257:474 Camelot Boulevard, Bath, WY 28356:5/13/69:24500

Minerva McGonagall:408-233-8971:45 Rose Terrace, San Francisco, CA 92303:2/3/36:25000

Zippy Pinhead:834-823-8319:2356 Bizarro Ave., Farmount, IL 84357:1/1/67:89500

aavbox26@Ubuntu1:/home$
```

After writing the command written above which was **sed -i -e '$d' SedLab,** we can see in the screenshot that the five lines were **removed.** The command deletes empty lines of text along with the lines containing **Luna Lovegood, Popeye Sailor, and finally Dorothy Gale.**

Sources:

https://www.grymoire.com/Unix/Sed.html#uh-30 **What is deleting in sed?**

https://www.theunixschool.com/2012/06/sed-25-examples-to-delete-line-or.html **Deleting command examples.**

Question **3.**

```
aavbox26@Ubuntu1:/mnt/sed$ sed -n '3,15p' SedLab
Betty Boop:245-836-8357:635 Cutesy Lane, Hollywood, CA 91464:6/23/23:14500

Dorian Gray:385-375-8395:3567 Populus Place, Caldwell, NJ 23875:6/18/68:23400

Holly Golightly:397-857-2735:74 Pine Street, Dearborn, MI 23874:3/28/45:245700

Ebenezer Scrooge:548-834-2348:583 Laurel Ave., Kingsville, TX 83745:10/1/35:58900

Hester Prynne:408-253-3122:123 Park St., San Jose, CA 04086:7/25/53:85100

Westley Pirate:284-758-2857:23 Edgecliff Place, Lincoln , NB 92743:7/25/53:85100

Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200
aavbox26@Ubuntu1:/mnt/sed$
```

The following command prints lines **3-15** in a file named **SedLab.** The command written was **sed -n '3,15p' SedLab.** There are **7 lines** with text, there are also **6 blank** lines that do not contain any text. The **-n** restricts the text lines that get printed within the file to the specified pattern, which would be a number range like **3-15.** The pattern in this command is the range from **3-15,** and **(p)** stands for **print** along with the line ending with the **filename** to search for the text lines within the file named **SedLab.**

Sources:

How to print, combined command with ranges.

https://www.grymoire.com/Unix/Sed.html#uh-9 **Print**

https://www.grymoire.com/Unix/Sed.html#uh-28 **Ranges.**

https://www.gnu.org/software/sed/manual/html_node/Command_002dLine-Options.html **Options for sed.**

Question **4.**

```
Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200

Lizzie Bennett:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900

Nancy Drew:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900

Jo March:327-832-5728:3465 Mirlo Street, Peabody, MA 34756:10/2/65:35200

Victor Frankenstein:835-365-1284:454 Easy Street, Decatur, IL 75732:2/28/53:123500

Ephram Hardy:293-259-5395:235 CarltonLane, Joliet, IL 73858:8/12/20:56700

Meg Murry:834-938-8376:23445 Aster Ave., Allentown, NJ 83745:12/1/38:45000

Lucy Pevensie:385-573-8326:832 Ponce Drive, Gzary, IN 83756:12/1/46:268500

Hemione Granger:408-456-1234:4 Harvard Square, Boston, MA 02133:4/22/62:52600

Mary Poppins:846-836-2837:6937 Ware Road, Milton, PA 93756:9/21/46:43500

Sir Lancelot:837-835-8257:474 Camelot Boulevard, Bath, WY 28356:5/13/69:24500


Zippy Pinhead:834-823-8319:2356 Bizarro Ave., Farmount, IL 84357:1/1/67:89500

Dorothy Gale:923-835-8745:23 Wimp Lane, Kensington, DL 38758:8/31/69:126000

Popeye Sailor:156-454-3322:945 Bluto Street, NotHere, USA 29358:3/19/35:22350

Luna Lovegood:385-898-8357:38 Fife Way, Abilene, TX 39673:1/5/58:95600


Molly Weasley:387-827-1095:13 Uno Lane, Ashville, NC 23556:7/1/29:57000

Arya Stark:438-910-7449:8235 Maple Street, Wilmington, VM 29085:9/23/63:68900
aavbox26@Ubuntu1:/mnt/sed$ sed '/CA/d' SedLab_
```

The following command deletes all instances of **CA** (California) from the file **SedLab** in the **Ubuntu** terminal. I opened the file **SedLab** in word, I verified there were **4 lines** for **CA.** In the **Ubuntu** terminal when writing the command **sed '/CA/d' SedLab, there are four lines that appear** to be empty and look double spaced for all the instances where **California** was deleted in the terminal. All lines that contained **CA** were removed, because of this, it looks like there are two empty lines of text right after each other. This is because the next line after each instance of **CA** were empty text lines.

In your own file for this command you would write **sed '/pattern/d' filename. d** in sed deletes an entire line, and proceeds to the next line.

Sources:

https://www.grymoire.com/Unix/Sed.html#uh-30 **Deleting.**

Question **5.**

```
aavbox26@Ubuntu1:/mnt/sed$ sed -n '/\/[1-7]\//p' SedLab
Ebenezer Scrooge:548-834-2348:583 Laurel Ave., Kingsville, TX 83745:10/1/35:58900
Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200
Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200
Jo March:327-832-5728:3465 Mirlo Street, Peabody, MA 34756:10/2/65:35200
Meg Murry:834-938-8376:23445 Aster Ave., Allentown, NJ 83745:12/1/38:45000
Lucy Pevensie:385-573-8326:832 Ponce Drive, Gzary, IN 83756:12/1/46:268500
Minerva McGonagall:408-233-8971:45 Rose Terrace, San Francisco, CA 92303:2/3/36:25000
Zippy Pinhead:834-823-8319:2356 Bizarro Ave., Farmount, IL 84357:1/1/67:89500
Luna Lovegood:385-898-8357:38 Fife Way, Abilene, TX 39673:1/5/58:95600
Molly Weasley:387-827-1095:13 Uno Lane, Ashville, NC 23556:7/1/29:57000
aavbox26@Ubuntu1:/mnt/sed$
```

The following command written for this example, **prints the birthdays that are within the first week of the month.** This command follows the **MM/DD/YY, month, date, year format.** The command written was,

sed -n '/\/[1-7]\//p' SedLab. Why does this command work? The command works because it is **printing the birthdays** that start within the **first week** of the month. The command uses **output, ranges, and slash as delimiters.** A **"Delimimiter"** is a character to separate the different parts of a command line.

-n and p were used in this command to have matching lines printed. The range of **[1-7]** was to find the day only, in the **month, date, and year format, MM/DD/YY** to ensure that the day was for the **first week of the month. Backslashes** are written to **escape the same character or quote the character** which were also **slashes**, so the sed system does not get confused with **similar characters.** Specifically the forward slash, so the pattern does not end before finishing. The **forward slashes (\/)** are **used for the date format**, and are treated as real characters for the search of a pattern.

According to grymoire, any type of delimiter works like a **slash**

(// \/ \\ ) or a **pipe (|)** to separate portions of the command. If you do not have three delimiters you will have an error like unterminated 's' command according to grymoire.

Sources:

Combining three commands.

https://www.grymoire.com/Unix/Sed.html#uh-15b **-n, p** command.

https://www.grymoire.com/Unix/Regular.html#uh-5 **Range. 1-7 month, 1-7 days, 1-7 year.**

https://www.grymoire.com/Unix/Sed.html#uh-2 **Slash** options.

Question **6.**

```
Lizzie Bennett:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900

Nancy Drew:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900

Jo March:327-832-5728:3465 Mirlo Street, Peabody, MA 34756:10/2/65:35200

Victor Frankenstein:835-365-1284:454 Easy Street, Decatur, IL 75732:2/28/53:123500

Ephram Hardy:293-259-5395:235 CarltonLane, Joliet, IL 73858:8/12/20:56700

Meg Murry:834-938-8376:23445 Aster Ave., Allentown, NJ 83745:12/1/38:45000

Lucy Pevensie:385-573-8326:832 Ponce Drive, Gzary, IN 83756:12/1/46:268500

Hemione Granger:408-456-1234:4 Harvard Square, Boston, MA 02133:4/22/62:52600

Mary Poppins:846-836-2837:6937 Ware Road, Milton, PA 93756:9/21/46:43500

Sir Lancelot:837-835-8257:474 Camelot Boulevard, Bath, WY 28356:5/13/69:24500 ***

Minerva McGonagall:408-233-8971:45 Rose Terrace, San Francisco, CA 92303:2/3/36:25000

Zippy Pinhead:834-823-8319:2356 Bizarro Ave., Farmount, IL 84357:1/1/67:89500

Dorothy Gale:923-835-8745:23 Wimp Lane, Kensington, DL 38758:8/31/69:126000

Popeye Sailor:156-454-3322:945 Bluto Street, NotHere, USA 29358:3/19/35:22350

Luna Lovegood:385-898-8357:38 Fife Way, Abilene, TX 39673:1/5/58:95600

Daenerys Targaryen:408-724-0140:1222 Oxbow Court, Sunnyvale, CA 94087:5/19/66:34200

Molly Weasley:387-827-1095:13 Uno Lane, Ashville, NC 23556:7/1/29:57000

Arya Stark:438-910-7449:8235 Maple Street, Wilmington, VM 29085:9/23/63:68900
aavbox26@Ubuntu1:/mnt/sed$ sed '/^Sir/ s/$/ ***/' SedLab_
```

The following command appends three asterisks **(***) to the end of lines starting with 'Sir'. sed '/^Sir/ s/$/ ***/'SedLab.** This command works by looking for the pattern written as **'Sir', with the caret looking for the pattern at the beginning of the line.** Once the pattern is found, **(s) for substitution** command

changes the line, **($) dollar sign** looks for the end of the line**,** the end of the line has blank text, it is then replaced with **asterisks (***).**

In the file **SedLab**, Sir Lancelot is the only line that begins with **'Sir'** in the text file. Because of this the three asterisks are added to the end of Sir Lancelot's line, it is the only line changed, while the other text lines are left **unchanged.**

Sources:

https://stackoverflow.com/questions/9913339/how-to-add-text-at-the-end-of-a-line-in-unix **Example** commands.

https://stackoverflow.com/questions/23078897/how-to-add-text-at-the-end-of-each-line-in-unix **Adding text to the end of a line.**

https://www.grymoire.com/Unix/Sed.html#uh-1 **Substitution command.**

https://www.grymoire.com/Unix/Regular.html#uh-2 **^ for pattern. $ end of line.**

Question **7.**

```
Lizzie Bennett:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900
Nancy Drew:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900
Jo March:327-832-5728:3465 Mirlo Street, Peabody, MA 34756:10/2/65:35200
Victor Frankenstein:835-365-1284:454 Easy Street, Decatur, IL 75732:2/28/53:123500
Ephram Hardy:293-259-5395:235 CarltonLane, Joliet, IL 73858:8/12/20:56700
Meg Murry:834-938-8376:23445 Aster Ave., Allentown, NJ 83745:12/1/38:45000
Lucy Pevensie:385-573-8326:832 Ponce Drive, Gzary, IN 83756:12/1/46:268500
Hemione Granger:408-456-1234:4 Harvard Square, Boston, MA 02133:4/22/62:52600
Mary Poppins:846-836-2837:6937 Ware Road, Milton, PA 93756:9/21/46:43500
Sir Lancelot:837-835-8257:474 Camelot Boulevard, Bath, WY 28356:5/13/69:24500
Minerva McGonagall:408-233-8971:45 Rose Terrace, San Francisco, CA 92303:2/3/36:25000
Zippy Pinhead:834-823-8319:2356 Bizarro Ave., Farmount, IL 84357:1/1/67:89500
Dorothy Gale:923-835-8745:23 Wimp Lane, Kensington, DL 38758:8/31/69:126000
Popeye Sailor:156-454-3322:945 Bluto Street, NotHere, USA 29358:3/19/35:22350
Luna Lovegood:385-898-8357:38 Fife Way, Abilene, TX 39673:1/5/58:95600
Daenerys Targaryen:408-724-0140:1222 Oxbow Court, Sunnyvale, CA 94087:5/19/66:34200
Molly Weasley:387-827-1095:13 Uno Lane, Ashville, NC 23556:7/1/29:57000
Arya Stark:438-910-7449:8235 Maple Street, Wilmington, VM 29085:9/23/63:68900
```

The first command is **sed '/Westley Pirate/c\As you wish.' SedLab.** This command replaces all instances of the person with the name 'Westley Pirate' to 'As you wish'. For your file you would write **sed '/Firstname Lastname/c\unique phrase.' Filename.**

```
Molly Weasley:387-827-1095:13 Uno Lane, Ashville, NC 23556:7/1/29:57000

Arya Stark:438-910-7449:8235 Maple Street, Wilmington, VM 29085:9/23/63:68900
aavbox26@Ubuntu1:/mnt/sed$ sed '/Westley Pirate/c\As you wish.' SedLab | grep "As you wish"
As you wish.
As you wish.
As you wish.
aavbox26@Ubuntu1:/mnt/sed$ _
```

The second command was a continuation of the previous command but adding a **pipe (|)** Along with **grep,** the pattern **"As you wish"** so that **sed** can read the output of the line that was changed, which was the **Westley Pirate** lines. **Grep** in this command, searches for the three lines changed; it acts as a quick search or tool to find the three lines. There are three different instances within the file where **Westley Pirate** appears. With **the (c) for change,** it changes all instances of Westley Pirate to 'As you wish'. Since **Westley Pirate appears 3 times,** there are three instances of **'As you wish'** in the terminal.

Sources:

https://www.grymoire.com/Unix/Sed.html#uh-39 **Adding, and changing lines command examples. The c command.**

https://unix.stackexchange.com/questions/24140/return-only-the-portion-of-a-line-after-a-matching-pattern **Example second command, | grep.**

https://superuser.com/questions/1107680/how-to-use-sed-with-piping **sed and piping examples.**

https://www.grymoire.com/Unix/Sed.html#uh-12 **Arguments and Invocation of sed, why use the pipe (|).**

Question **8.**

```
Molly Weasley:387-827-1095:13 Uno Lane, Ashville, NC 23556:7/1/29:57000

Arya Stark:438-910-7449:8235 Maple Street, Wilmington, VM 29085:9/23/63:68900
aavbox26@Ubuntu1:/mnt/sed$ sed '/Minerva McGonagall/s|[0-9]*/[0-9]*/[0-9]*|10/04/1935|' SedLab | gre
p "Minerva"
Minerva McGonagall:408-233-8971:45 Rose Terrace, San Francisco, CA 92303:10/04/1935:25000
aavbox26@Ubuntu1:/mnt/sed$ _
```

Changing **'Minerva McGonagall's'** birthday requires a regular expression. In this example, the regular expression written was…

sed '/Minerva McGonagall/s|[0-9]*/*[0-9]*/[0-9]*|10/04/1935|' SedLab .

This command uses a **regular expression** to find the existing birthdate and changes it. The **pipe** in this command is used as a **delimiter.** A **"Delimimiter"** is a character to separate the different parts of a command line. Pipe is usually used to search for two patterns at once in a command.

For your file the command should be written as follows…

sed '/Firstname Lastname/s|[0-9]*/*[0-9]*/[0-9]*|10/04/1935|' Filename .

To explain this further and breakdown the command, sed is the program we are working with so we start the command line with **sed.** Minerva McGonagall is the **pattern/sequence of characters** within the command, the **slashes are delimiters** or also known as **escape characters** so that sed is not confused with looking for other slashes in the command. **/s/ means substitution** command, **pipe lets you search for two patterns at once. [0-9] is the range** and matches digits starting from **0-9.**

The **forward slashes (/)** are used as characters within the command to match the slashes that are used within the file for the **MM/DD/YY** format. Finally the **asterisks (*)** are written within the command because they match any number of digits within the command, in this case **one to two digits within the three slashes.** Within a regular expression command, the **asterisks look for zero or more characters.** It looks for a **one digit month or two digit month.** The asterisk makes

sure that the month or day is between **one or two digits,** and the command will search for it.

When reading the file in word, the **SedLab file containing 53 lines** you can see that there is **one instance** of a person named **'Minerva McGonagall'.** I only saw one instance, but there could be more. Minerva's original birthday within the file is **2/3/36.** For this question, the person's birthdate was intended to be changed from **2/3/36** to **10/04/1935.** For the birthdate change for **Minerva** this command uses the **MM/DD/YY format.**

The second command was a **continuation of the original command**. This command uses **pipe (|) grep pattern- ("Minerva")** to see if the birthdate for the individual person and line of text was **changed from 2/3/36 to 10/04/1935.**

Second command below....

sed '/Minerva McGonagall/s|[0-9]*/*[0-9]*/[0-9]*|10/04/1935|' SedLab | grep "Minerva".

Grep in this command, it acts as a quick search or tool to find the name/pattern **"Minerva".**

For your file the command should be written as follows,

sed '/Firstname Lastname/s|[0-9]*/*[0-9]*/[0-9]*|10/04/1935|' Filename | grep "Firstname".

Grep in this command, it acts as a quick search or tool to find the name/pattern **"Firstname"** in your file.

Sources:

https://www.grymoire.com/Unix/Sed.html#uh-1 **Substitution command.**

https://www.grymoire.com/Unix/Regular.html#uh-6 **Character Sets.**

https://www.grymoire.com/Unix/Regular.html#uh-7 **Repeating characters sets with *.**

https://www.grymoire.com/Unix/Sed.html#uh-12 Arguments and Invocation of sed, why use the pipe (|).

https://www.grymoire.com/Unix/Sed.html#uh-2 Slash options and examples.

https://web.archive.org/web/20220816210438/https://www.grymoire.com/Unix/Sed.html#uh-22 **Passing arguments in sed.**

https://www.grymoire.com/Unix/Grep.html#uh-4 Grep as a filter.

Question **9.**

```
aavbox26@Ubuntu1:/mnt/sed$ sed '/^$/d' SedLab
Huckleberry Finn:238-923-7366:95 Latham Lane, Easton, PA 83755:11/12/56:20300
Betty Boop:245-836-8357:635 Cutesy Lane, Hollywood, CA 91464:6/23/23:14500
Dorian Gray:385-375-8395:3567 Populus Place, Caldwell, NJ 23875:6/18/68:23400
Holly Golightly:397-857-2735:74 Pine Street, Dearborn, MI 23874:3/28/45:245700
Ebenezer Scrooge:548-834-2348:583 Laurel Ave., Kingsville, TX 83745:10/1/35:58900
Hester Prynne:408-253-3122:123 Park St., San Jose, CA 04086:7/25/53:85100
Westley Pirate:284-758-2857:23 Edgecliff Place, Lincoln , NB 92743:7/25/53:85100
Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200
Westley Pirate:284-758-2867:23 Edgecliff Place, Lincoln, NB 92743:11/3/35:58200
Lizzie Bennett:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900
Nancy Drew:674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900
Jo March:327-832-5728:3465 Mirlo Street, Peabody, MA 34756:10/2/65:35200
Victor Frankenstein:835-365-1284:454 Easy Street, Decatur, IL 75732:2/28/53:123500
Ephram Hardy:293-259-5395:235 CarltonLane, Joliet, IL 73858:8/12/20:56700
Meg Murry:834-938-8376:23445 Aster Ave., Allentown, NJ 83745:12/1/38:45000
Lucy Pevensie:385-573-8326:832 Ponce Drive, Gzary, IN 83756:12/1/46:268500
Hemione Granger:408-456-1234:4 Harvard Square, Boston, MA 02133:4/22/62:52600
Mary Poppins:846-836-2837:6937 Ware Road, Milton, PA 93756:9/21/46:43500
Sir Lancelot:837-835-8257:474 Camelot Boulevard, Bath, WY 28356:5/13/69:24500
Minerva McGonagall:408-233-8971:45 Rose Terrace, San Francisco, CA 92303:2/3/36:25000
Zippy Pinhead:834-823-8319:2356 Bizarro Ave., Farmount, IL 84357:1/1/67:89500
Dorothy Gale:923-835-8745:23 Wimp Lane, Kensington, DL 38758:8/31/69:126000
Popeye Sailor:156-454-3322:945 Bluto Street, NotHere, USA 29358:3/19/35:22350
Luna Lovegood:385-898-8357:38 Fife Way, Abilene, TX 39673:1/5/58:95600
Daenerys Targaryen:408-724-0140:1222 Oxbow Court, Sunnyvale, CA 94087:5/19/66:34200
Molly Weasley:387-827-1095:13 Uno Lane, Ashville, NC 23556:7/1/29:57000
Arya Stark:438-910-7449:8235 Maple Street, Wilmington, VM 29085:9/23/63:68900
aavbox26@Ubuntu1:/mnt/sed$ _
```

The following command for the screenshot above has multiple parts to get it to work properly. If you wanted to delete all the empty text lines from the file "SedLab" or your own file named **(placeholder name)-"Filename"**, you would start with **sed** the

delimiter characters like **slashes (/ /)** for the command to read like **MM/DD/YY** format.

In between the delimiter characters, the slashes **(/ /)** there are anchor characters. Anchor characters are used to find the beginning or an end of a pattern for a specific text line, or line number. According to the grymoire source, the **(^) caret character** is used as a starting point, while the **($) dollar sign character** is used as an ending anchor character. The final part of this command is **(d)** which stands for **delete character**, this deletes the empty text lines within the **'Filename'** or **SedLab file.**

Sources:

Combining two anchor characters **and the** delete command **to get the intended result.**

https://www.grymoire.com/Unix/Regular.html#uh-2 (^) Beginning Character, ($) Ending Character Examples.

https://www.grymoire.com/Unix/Sed.html#uh-30 Deleting command.

Question **10.**

```
  GNU nano 6.2                                  char.sed *
1i Great Literary Characters
s/\([0-9]\{3\}-[0-9]\{3\}-[0-9]\{4\}\)/1+\1/
$a Happily Ever after. The End.
```

Nano char.sed, creates the text file within the **Ubuntu, sed** environment. What is written in the text file named **char.sed**, short for "**characters.sed"**, are three different commands. The first is the **1i command. 1i** means **insert at line 1. s/** is for substitution, **$a command** means append to the end of the file.

Then following a pattern named **"Great Literary Characters".** We would then write out this command, **s/\([0-9]\{3\}-[0-9]\{3\}-[0-9]\{4\}\)/1+\1/ .** This command is a **substitution command**, it is a command that looks for and edits the **phone numbers** for each line.

This command is built by combining different commands together. Substitute command, **s /old / new/**, character set **0-9**, matching a specific number of sets **[a-z]\{4,8} or {3\} {4\}** interval expressions. **Slashes** were used as **delimiters**, because the phone numbers contain **dashes (-)** this would be hard to read in a command. **/1+** is the change to the phone numbers, **\1/**is a **backreference** and remembers the patterns.

Finally after writing the second command, you would append to the end of the line, at the end of the line there would be a pattern, the text would be **"Happily Ever after. The End."**

Next we would save the text file named **char.sed.** Or in your case you would write textfilename.sed, using **ctrl + O** and then exiting the text editor with **ctrl + X.** In the terminal we would write **sed -f char.sed Sedlab**, or in your case **sed -f textfilename.sed Filename,** to see the results of the commands written within the text file in our **SedLab** file or **Filename. -f** means **file,** it combines the **text file script char.sed with the SedLab file text lines.**

```
Lizzie Bennett:1+674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900

Nancy Drew:1+674-843-1385:20 Parak Lane, Duluth, MN 23850:4/12/23:780900

Jo March:1+327-832-5728:3465 Mirlo Street, Peabody, MA 34756:10/2/65:35200

Victor Frankenstein:1+835-365-1284:454 Easy Street, Decatur, IL 75732:2/28/53:123500

Ephram Hardy:1+293-259-5395:235 CarltonLane, Joliet, IL 73858:8/12/20:56700

Meg Murry:1+834-938-8376:23445 Aster Ave., Allentown, NJ 83745:12/1/38:45000

Lucy Pevensie:1+385-573-8326:832 Ponce Drive, Gzary, IN 83756:12/1/46:268500

Hemione Granger:1+408-456-1234:4 Harvard Square, Boston, MA 02133:4/22/62:52600

Mary Poppins:1+846-836-2837:6937 Ware Road, Milton, PA 93756:9/21/46:43500

Sir Lancelot:1+837-835-8257:474 Camelot Boulevard, Bath, WY 28356:5/13/69:24500

Minerva McGonagall:1+408-233-8971:45 Rose Terrace, San Francisco, CA 92303:2/3/36:25000

Zippy Pinhead:1+834-823-8319:2356 Bizarro Ave., Farmount, IL 84357:1/1/67:89500

Dorothy Gale:1+923-835-8745:23 Wimp Lane, Kensington, DL 38758:8/31/69:126000

Popeye Sailor:1+156-454-3322:945 Bluto Street, NotHere, USA 29358:3/19/35:22350

Luna Lovegood:1+385-898-8357:38 Fife Way, Abilene, TX 39673:1/5/58:95600

Daenerys Targaryen:1+408-724-0140:1222 Oxbow Court, Sunnyvale, CA 94087:5/19/66:34200

Molly Weasley:1+387-827-1095:13 Uno Lane, Ashville, NC 23556:7/1/29:57000

Arya Stark:1+438-910-7449:8235 Maple Street, Wilmington, VM 29085:9/23/63:68900
Happily Ever after. The End.
aavbox26@Ubuntu1:/mnt/sed$ sed -f char.sed SedLab
```

Once the script is saved, we check if the text file **char.sed** is in the same **folder/ directory** as the **SedLab** file. Running this command, **sed -f char.sed SedLab.** You would know that they are now in the same file and directory when the text **"Great Literary Characters"** appears at the beginning of the file and the text "Happily Ever after. The End" shows up at the end of the file named "**SedLab" or your file named "Filename".**

```
aavbox26@Ubuntu1:/mnt/sed$ sed -f char.sed SedLab | head -n 5
Great Literary Characters
Huckleberry Finn:1+238-923-7366:95 Latham Lane, Easton, PA 83755:11/12/56:20300

Betty Boop:1+245-836-8357:635 Cutesy Lane, Hollywood, CA 91464:6/23/23:14500

aavbox26@Ubuntu1:/mnt/sed$ _
```

To find the beginning text lines of the file **SedLab** with the changes from **char.sed** text file, we would write out the **head** command. The command in the example is written like this, **sed -f char.sed SedLab | head -n 5.** For your file you would write **sed -f textfilename.sed Filename | (pipe) head -n 5.**

-f means file, **sed is written twice**, the first instance of **sed** in the line is the type of program you are in, the second is the **.sed** and this is considered a file extension. **Pipe (|)** is an **operator,** it allows different commands to be combined. The **head command** is used to display the first few lines of a file like **Sedlab or your filename,** the number next to it is the **amount of lines** you want to be read from the file.

For this example I used **5 as I wanted to see the first five lines printed.** There is a limit, the limit is usually up to **10 lines by default.** There can be a higher limit for the head command, based on how many lines of text you have within a file like **"SedLab"** or your **"Filename".**

Final Sources:

https://unix.stackexchange.com/questions/95939/how-exactly-do-i-create-a-sed-script-and-use-it-to-edit-a-file

I was very confused with how to create one, but this source was a good foundation on creating a script for sed. Never written this type of script before, only worked with simple bash scripts.

https://pressbooks.senecapolytechnic.ca/uli101/chapter/text-editors-in-linux/ **Nano Text. nano char.sed.**

https://stackoverflow.com/questions/19164976/how-to-use-sed-to-change-file-extensions **What is a file extension in sed and Linux? .sed .txt etc.**

https://unix.stackexchange.com/questions/99350/how-to-insert-text-before-the-first-line-of-a-file **Characters at the beginning of the file examples. 1i etc.**

https://www.grymoire.com/Unix/Sed.html#uh-41 **Inserting a line.**

https://www.grymoire.com/Unix/Sed.html#uh-1 **Substitute command.**

https://www.grymoire.com/Unix/Regular.html#uh-6 **Character Sets.**

https://www.grymoire.com/Unix/Regular.html#uh-7 **Matching a specific number of sets.**

https://www.grymoire.com/Unix/Sed.html#uh-40 **Appending a line.**

https://www.grymoire.com/Unix/Sed.html#uh-39 **Adding, Changing, Inserting new lines in sed examples.**

https://www.grymoire.com/Unix/Sed.html#uh-12 **Arguments and Invocation of sed,** why use the pipe (|).

https://superuser.com/questions/1745781/is-pipe-a-command **What is pipe?**

https://www.geeksforgeeks.org/linux-unix/piping-in-unix-or-linux/ **More about Piping.**

https://www.geeksforgeeks.org/linux-unix/head-command-linux-examples/ **Head command examples.**

https://www.grymoire.com/Unix/Regular.html#uh-10 **Back references.**